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Abstract. Einstein velocity addition endows the unit ball of $\mathbb{R}^n$ with the structure of a gyrocommutative gyrogroup together with the hyperbolic Hilbert–Klein Riemannian metric. First, we propose a self-contained derivation of these properties by rewriting Einstein addition in the Clifford algebra of the Euclidean space $\mathbb{R}^n$, in which gyroautomorphisms are given by spinors. Then, our main result shows that the Einstein gyrogroup arises naturally in the context of quantum measurements in real spin factors. By embedding the state space of the Jordan algebra $\mathbb{R} \oplus \mathbb{R}^n$ into a multipartite rebit system realized as a tensor product of the algebra of 2×2 real symmetric matrices, we prove that Lüders state-update operations induce a gyrogroup morphism from the unit ball endowed with Einstein addition to a gyrogroup of density matrices. This extends known results in the low-dimensional case for the rebit and the qubit, and confirms that relativistic velocities and quantum state-updates in real spin factor share a common algebraic structure in arbitrary dimensions.

Keywords: *Einstein gyrogroup, Spin factor, Quantum information, Relativistic velocity addition, Clifford algebra, Jordan algebra, Hilbert-Klein metric*

1. **Introduction**

The addition of relativistic velocities, arising from the work of Einstein and Poincaré, provides a fundamental and subtle example of a binary operation that is neither commutative nor associative (see, e.g., [7]). When expressed in the unit ball of $\mathbb{R}^n$, this operation gives rise to a rich algebraic structure extensively studied by Ungar through the theory of gyrogroups (see, e.g., [22]). Although Einstein addition is known to satisfy the axioms of a gyrocommutative gyrogroup, the existing derivations either rely on computer algebra (see for instance Definition 3.45, p. 88 in [22]) or invoke results from B-loop theory [15], [19].

This motivates us to present in Section 2 a more self-contained, simple proof inspired by [6], which addressed the simpler case of Möbius addition. We rewrite Einstein addition within the Clifford algebra of the Euclidean space $\mathbb{R}^n$, thereby allowing us to naturally realize Einstein gyroautomorphisms as spinors. From a geometric viewpoint, we also recall how the Einstein gyrogroup naturally endows the unit ball of $\mathbb{R}^n$, for all $n \geq 2$, with the hyperbolic Hilbert-Klein Riemannian metric.

While Section 2 summarizes the fundamental properties of the Einstein gyrogroup, together with proofs accessible to readers not familiar with gyrogroup and B-loop theories, the main aim of this work is to prove that the Einstein gyrogroup is not only the algebraic structure governing relativistic velocity composition, but also arises intrinsically in the description of quantum measurements on real spin factor state spaces.

Spin factors are rank-two formally real Jordan algebras that provide an algebraic alternative to the Hilbert-space formulation of quantum theory. The real spin factor $\mathbb{R} \oplus \mathbb{R}^2$ corresponds to the so-called rebit, whose state space is the unit disk, while the complex spin factor $\mathbb{C} \oplus \mathbb{C}^2$ yields the qubit with the Bloch ball as its state space (see [2]). In these models, quantum measurements are described by Lüders operations, which act on density matrices and determine both expectation values and post-measurement states.

For the qubit, connections between post-measurement states and Einstein velocity addition have been observed in several works [1, 4, 5, 14], and related observations were made for the rebit case in [3]. However, these results rely on the special low-dimensional structure of 2×2 matrices and do not extend directly to arbitrary spin factors. In particular, up to our knowledge, there exists no work devoted to explain the appearance of Einstein addition for real spin factors $\mathbb{R} \oplus \mathbb{R}^n$ with arbitrary $n \geq 2$.

The main result of this article fills this gap. After reviewing spin factors, we explain in Section 3 how the unit ball of $\mathbb{R}^n$ embeds into the state space of the spin factor $\mathbb{R} \oplus \mathbb{R}^n$, identified with a section of the future light cone in $(n+1)$-dimensional Minkowski space. Then, following [10], we construct an embedding of this spin factor into a multipartite rebit system, realized as a tensor product of the algebra of 2×2 real symmetric matrices.

Finally, in Section 4, we derive the main theorem of this work (Theorem 4.3). We prove that Lüders operations performed on density matrices of this multipartite rebit system induce a gyrogroup morphism

$$\chi \circ \Sigma : (B^n, \oplus) \longrightarrow (\mathcal{D} = \chi(\mathcal{S}(\mathbb{R} \oplus \mathbb{R}^n)), \oplus) \tag{1}$$

between the unit ball of $\mathbb{R}^n$ equipped with Einstein addition $(\mathbf{e}, \mathbf{s}) \longmapsto \mathbf{e} \oplus \mathbf{s}$ and the gyrogroup of density matrices whose gyrooperation is given by the state-update operation

$$\rho_{\mathbf{e}} \oplus \rho_{\mathbf{s}} = \frac{2\rho_{\mathbf{e}}^{1/2} \rho_{\mathbf{s}} \rho_{\mathbf{e}}^{1/2}}{1 + \langle \mathbf{e} | \mathbf{s} \rangle} = \rho_{\mathbf{e} \oplus \mathbf{s}}. \tag{2}$$

Note that rather than arising from a change of coordinates between two models of the ball, the gyrogroup law emerges intrinsically from the properties of Lüders operations. Thus, this theorem establishes a direct algebraic bridge between relativistic velocity addition and quantum measurements in real spin factors of arbitrary dimensions.

2. Einstein gyrogroup

The algebraic structure of gyrogroups has been extensively investigated by Ungar (see, for instance, [22] and the references therein). Two fundamental examples are provided by the Möbius and Einstein gyrogroups. In Ungar's work, however, the verification that Möbius and Einstein addition satisfy the axioms of a gyrocommutative gyrogroup is largely carried out with the aid of computer algebra, rather than through fully explicit handwritten analytical proofs (see, e.g., Definition 3.45, p. 88 in [22]). A Clifford algebra approach leading to explicit proofs was first developed in [6] for the Möbius gyrogroup, and later combined with results from the theory of B-loops to obtain an algebraic proof for the Einstein gyrogroup (see [15], [19]).

Here, we provide a more self-contained, simple proof that Einstein addition endows the unit ball of $\mathbb{R}^n$ with the structure of a gyrocommutative gyrogroup. Unlike previous approaches, we consider $\mathbb{R}^n$ embedded into the Clifford algebra associated with the Euclidean space (of positive signature) rather than one of negative signature.

2.1. Notations and definitions.

Definition 2.1. A magma $(G, \oplus)$ is a gyrocommutative gyrogroup if its binary operation $\oplus$ satisfies

1. There exists a left identity element 0 of G such that $0 \oplus a = a$ for all a in G.
2. Every element a of G admits a left inverse $\ominus a$ in G, with $\ominus a \oplus a = 0$.
3. For all elements a and b of G, there exists a gyroautomorphism $G(a, b)$ of the magma$(G, \oplus)$ satisfying
   - P1: $a \oplus (b \oplus c) = (a \oplus b) \oplus G(a, b)(c)$ for all elements a, b, and c of G (gyroassociativity).
   - P2: $G(a, b) = G(a \oplus b, b)$ for all elements a and b of G (left loop property).
   - P3: $a \oplus b = G(a, b)(b \oplus a)$ for all elements a and b of G (gyrocommutativity). •

A gyrocommutative gyrogroup whose gyroautomorphisms $G(a, b)$ are the identity for all a and b, is a commutative group. The main goal of this section is to prove the following theorem.

Theorem 2.2. *The ball $B^n = \{\boldsymbol{s} \in \mathbb{R}^n, \|\boldsymbol{s}\| \leq 1\}$ endowed with the Einstein addition defined by*

$$\boldsymbol{s} \oplus \boldsymbol{t} = \frac{1}{1+\langle \boldsymbol{s}|\boldsymbol{t}\rangle}\left[\boldsymbol{s} + \frac{\boldsymbol{t}}{\gamma_{\boldsymbol{s}}} + \frac{\gamma_{\boldsymbol{s}}\langle \boldsymbol{s}|\boldsymbol{t}\rangle \boldsymbol{s}}{1+\gamma_{\boldsymbol{s}}}\right], \tag{3}$$

where

$$\gamma_{\boldsymbol{s}} = \frac{1}{\sqrt{1-\|\boldsymbol{s}\|^2}}, \tag{4}$$

if $\|\boldsymbol{s}\| < 1$, and

$$\boldsymbol{s} \oplus \boldsymbol{t} = \boldsymbol{s}, \tag{5}$$

if $\|\boldsymbol{s}\| = 1$, is a gyrocommutative gyrogroup. •

We consider $\mathbb{R}^n$ embedded in the Clifford algebra $\mathbb{R}_{n,0}$, with the quadratic form given by the Euclidean scalar product (see [8], [18]). Recall that there exists a unique involutive automorphism α of $\mathbb{R}_{n,0}$ defined by

$$\alpha(1) = 1, \quad \alpha(\mathbf{a}) = -\mathbf{a}, \quad \alpha(\mathbf{a}_{i_1}\mathbf{a}_{i_2}\cdots\mathbf{a}_{i_k}) = (-1)^k\mathbf{a}_{i_1}\mathbf{a}_{i_2}\cdots\mathbf{a}_{i_k}, \tag{6}$$

and a unique involutive antiautomorphism β of $\mathbb{R}_{n,0}$ defined by

$$\beta(1) = 1, \quad \beta(\mathbf{a}) = \mathbf{a}, \quad \beta(\mathbf{a}_{i_1}\mathbf{a}_{i_2}\cdots\mathbf{a}_{i_k}) = \mathbf{a}_{i_k}\mathbf{a}_{i_{k-1}}\cdots\mathbf{a}_{i_1}, \tag{7}$$

for all $\mathbf{a}$, $\mathbf{a}_{i_1}$, $\mathbf{a}_{i_2}$, ...,$\mathbf{a}_{i_k}$, in $\mathbb{R}^n$. This allows to define the "norm" $N(x)$ of an element x of $\mathbb{R}_{n,0}$ by

$$N(x) = x\overline{x} \tag{8}$$

with $\overline{x} = \beta \circ \alpha(x)$. Note that $N(\mathbf{a}) = -\|\mathbf{a}\|^2$ for all $\mathbf{a}$ in $\mathbb{R}^n$, where $\|\mathbf{a}\|$ is the Euclidean norm of $\mathbf{a}$. The inverse of an invertible element x, *i.e.* such that $N(x) \neq 0$, of $\mathbb{R}_{n,0}$ is

$$x^{-1} = \frac{\overline{x}}{N(x)}. \tag{9}$$

For $\mathbf{s}$ in B^n, we define

$$\delta_{\mathbf{s}} = \frac{\gamma_{\mathbf{s}}}{1+\gamma_{\mathbf{s}}}, \tag{10}$$

which implies

$$\delta_{\mathbf{s}}^2\|\mathbf{s}\|^2 = 2\delta_{\mathbf{s}} - 1. \tag{11}$$

With these notations,

$$\mathbf{s} \oplus \mathbf{t} = \frac{1}{1+\langle \mathbf{s}|\mathbf{t}\rangle}\left[(1+\delta_{\mathbf{s}}\langle \mathbf{s}|\mathbf{t}\rangle)\mathbf{s} + \frac{\mathbf{t}}{\gamma_{\mathbf{s}}}\right]. \tag{12}$$

2.2. **Proof of Theorem 2.2.** We begin by rewriting Einstein addition in the Clifford algebra $\mathbb{R}_{n,0}$; see [19] for a similar result in the Clifford algebra with negative signature.

Proposition 2.3. *Given $\boldsymbol{s}$ and $\boldsymbol{t}$ in B^n the Einstein addition $\boldsymbol{s} \oplus \boldsymbol{t}$ is defined by*

$$\boldsymbol{s} \oplus \boldsymbol{t} = \frac{2(\delta_{\boldsymbol{s}}\boldsymbol{s} + \delta_{\boldsymbol{t}}\boldsymbol{t})(1+\delta_{\boldsymbol{s}}\delta_{\boldsymbol{t}}\boldsymbol{t}\boldsymbol{s})}{N(1+\delta_{\boldsymbol{s}}\delta_{\boldsymbol{t}}\boldsymbol{s}\boldsymbol{t}) - N(\delta_{\boldsymbol{s}}\boldsymbol{s} + \delta_{\boldsymbol{t}}\boldsymbol{t})}. \tag{13}$$

Proof. Recalling that $\mathbf{st} = \langle \mathbf{s}|\mathbf{t}\rangle + \mathbf{s} \wedge \mathbf{t}$, and consequently $\mathbf{st} = 2\langle \mathbf{s}|\mathbf{t}\rangle - \mathbf{ts}$, we have

$$2(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t})(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}) = 4\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s} + 4\delta_{\mathbf{t}}(1-\delta_{\mathbf{s}})\mathbf{t} + 4\delta_{\mathbf{s}}^2\delta_{\mathbf{t}}\langle \mathbf{s}|\mathbf{t}\rangle\mathbf{s}. \tag{14}$$

Furthermore,

$$N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st}) = (1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}) = 1 + 2\delta_{\mathbf{s}}\delta_{\mathbf{t}}\langle \mathbf{s}|\mathbf{t}\rangle + \delta_{\mathbf{s}}^2\delta_{\mathbf{t}}^2\|\mathbf{s}\|^2\|\mathbf{t}\|^2, \tag{15}$$

and

$$N(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}) = -(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t})(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}) = -\delta_{\mathbf{s}}^2\|\mathbf{s}\|^2 - \delta_{\mathbf{t}}^2\|\mathbf{t}\|^2 - 2\delta_{\mathbf{s}}\delta_{\mathbf{t}}\langle \mathbf{s}|\mathbf{t}\rangle. \tag{16}$$

Consequently,

$$N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st}) - N(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}) = 4\delta_{\mathbf{s}}\delta_{\mathbf{t}}(1+\langle \mathbf{s}|\mathbf{t}\rangle). \tag{17}$$

By Cauchy-Schwarz inequality,

$$N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}) - N(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t}) \geq 4\delta_{\mathbf{s}}\delta_{\mathbf{t}}(1-\|\mathbf{s}\|\|\mathbf{t}\|) \geq 0, \tag{18}$$

with equality only if $\|\mathbf{s}\| = \|\mathbf{t}\| = 1$. Note that, in case of equality, one has $\delta_{\mathbf{s}} = \delta_{\mathbf{t}} = 1$, which gives

$$\frac{4\mathbf{s} + 4\langle\mathbf{s}|\mathbf{t}\rangle\mathbf{s}}{4(1+\langle\mathbf{s}|\mathbf{t}\rangle)} = \mathbf{s} = \mathbf{s} \oplus \mathbf{t}. \tag{19}$$

Finally, in all cases

$$\frac{2(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{t}\mathbf{s})}{N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}) - N(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})} = \frac{4\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s} + 4\delta_{\mathbf{t}}(1-\delta_{\mathbf{s}})\mathbf{t} + 4\delta_{\mathbf{s}}^2\delta_{\mathbf{t}}\langle\mathbf{s}|\mathbf{t}\rangle\mathbf{s}}{4\delta_{\mathbf{s}}\delta_{\mathbf{t}}(1+\langle\mathbf{s}|\mathbf{t}\rangle)} = \mathbf{s} \oplus \mathbf{t}. \bullet \tag{20}$$

Since

$$N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}) = N(1+\delta_{\mathbf{t}}\delta_{\mathbf{s}}\mathbf{t}\mathbf{s}), \tag{21}$$

one can check that

$$\|\mathbf{s}\oplus\mathbf{t}\|^2 = -\,(\mathbf{s}\oplus\mathbf{t})\overline{(\mathbf{s}\oplus\mathbf{t})} = \frac{4(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t})(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{t}\mathbf{s})(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})}{(N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}) - N(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t}))^2} \tag{22}$$

$$= -\,(\mathbf{t}\oplus\mathbf{s})\overline{(\mathbf{t}\oplus\mathbf{s})} = \|\mathbf{t}\oplus\mathbf{s}\|^2, \tag{23}$$

so that there exists a rotation $G(\mathbf{s},\mathbf{t})$ of $\mathbb{R}^n$ acting in the plane generated by $\mathbf{s}$ and $\mathbf{t}$ such that

$$\mathbf{s}\oplus\mathbf{t} = G(\mathbf{s},\mathbf{t})(\mathbf{t}\oplus\mathbf{s}). \tag{24}$$

This rotation is precisely the gyroautomorphism giving the gyrocommutativity (Property P3).

To verify the remaining properties P1 and P2, it is useful to identify the gyroautomorphism $G(\mathbf{s},\mathbf{t})$ with a spinor of the Clifford algebra $\mathbb{R}_{n,0}$ by means of the spin representation that describes the spin group $Spin(n,0)$ as a (universal) 2-sheeted covering of the special orthogonal group $SO(n,\mathbb{R})$ (see, e.g., [8]). This latter representation is given by

$$Spin(n,0) \longrightarrow SO(n,\mathbb{R}) \tag{25}$$

$$\tau \longmapsto (\mathbf{u} \longmapsto \tau\mathbf{u}\tau^{-1}). \tag{26}$$

Lemma 2.4. *For $\boldsymbol{s}$ and $\boldsymbol{t}$ in B^n,*

$$\boldsymbol{s}\oplus\boldsymbol{t} = G(\boldsymbol{s},\boldsymbol{t})(\boldsymbol{t}\oplus\boldsymbol{s}) = \tau(\boldsymbol{s},\boldsymbol{t})(\boldsymbol{t}\oplus\boldsymbol{s})\tau(\boldsymbol{s},\boldsymbol{t})^{-1}, \tag{27}$$

where the spinor $\tau(\boldsymbol{s},\boldsymbol{t}) \in Spin(n,0)$ of $\mathbb{R}_{n,0}$ writes

$$\tau(\boldsymbol{s},\boldsymbol{t}) = \frac{1+\delta_s\delta_t\boldsymbol{s}\boldsymbol{t}}{(N(1+\delta_s\delta_t\boldsymbol{s}\boldsymbol{t}))^{\frac{1}{2}}}. \tag{28}$$

Proof. Recalling that $N(\tau(\mathbf{s},\mathbf{t})) = 1$, we have

(29)

$$\tau(\mathbf{s},\mathbf{t})(\mathbf{t}\oplus\mathbf{s})\tau(\mathbf{s},\mathbf{t})^{-1} = \left[\frac{1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}}{(N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}))^{\frac{1}{2}}}\right]\left[\frac{2(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t})}{N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{t}\mathbf{s}) - N(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})}\right]\left[\frac{1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{t}\mathbf{s}}{(N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}))^{\frac{1}{2}}}\right]$$

$$= \frac{2(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t})(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})}{N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{t}\mathbf{s}) - N(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})} = \frac{2(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{t}\mathbf{s})}{N(1+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{s}\mathbf{t}) - N(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t})} \tag{30}$$

$$= \mathbf{s}\oplus\mathbf{t}. \bullet \tag{31}$$

Verifying Property P2 requires computing $\gamma_{\mathbf{s}\oplus\mathbf{t}}$ and $\delta_{\mathbf{s}\oplus\mathbf{t}}$.

Lemma 2.5. *For $\boldsymbol{s}$ and $\boldsymbol{t}$ in B^n, the constants $\gamma_{\boldsymbol{s}\oplus\boldsymbol{t}}$ and $\delta_{\boldsymbol{s}\oplus\boldsymbol{t}}$ are given by*

$$\gamma_{\boldsymbol{s}\oplus\boldsymbol{t}} = \frac{N(1+\delta_s\delta_t\boldsymbol{s}\boldsymbol{t}) - N(\delta_s\boldsymbol{s}+\delta_t\boldsymbol{t})}{N(1+\delta_s\delta_t\boldsymbol{s}\boldsymbol{t}) + N(\delta_s\boldsymbol{s}+\delta_t\boldsymbol{t})} = \gamma_s\gamma_t(1+\langle\boldsymbol{s}|\boldsymbol{t}\rangle), \tag{32}$$

and

$$\delta_{\boldsymbol{s}\oplus\boldsymbol{t}} = \frac{N(1+\delta_s\delta_t\boldsymbol{s}\boldsymbol{t}) - N(\delta_s\boldsymbol{s}+\delta_t\boldsymbol{t})}{2N(1+\delta_s\delta_t\boldsymbol{s}\boldsymbol{t})} = \frac{\delta_s\delta_t(1+\langle\boldsymbol{s}|\boldsymbol{t}\rangle)}{(1-\delta_s)(1-\delta_t)+\delta_s\delta_t(1+\langle\boldsymbol{s}|\boldsymbol{t}\rangle)}. \tag{33}$$

Proof. We have

$$1-\|\mathbf{s}\oplus\mathbf{t}\|^2 = 1+N(\mathbf{s}\oplus\mathbf{t}) = 1+(\mathbf{s}\oplus\mathbf{t})\overline{(\mathbf{s}\oplus\mathbf{t})} \tag{34}$$

$$= 1+\frac{4N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t})}{(N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})-N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t}))^2} = \frac{(N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})+N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t}))^2}{(N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})-N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t}))^2}. \tag{35}$$

Consequently, with Eq.(15) and Eq.(16),

$$\gamma_{\mathbf{s}\oplus\mathbf{t}} = \frac{N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})-N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t})}{N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})+N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t})} = \frac{\delta_\mathbf{s}\delta_\mathbf{t}(1+\delta_\mathbf{t}\langle\mathbf{s}|\mathbf{t}\rangle)}{(1-\delta_\mathbf{s})(1-\delta_\mathbf{t})}. \tag{36}$$

Since

$$\frac{\delta_\mathbf{s}\delta_\mathbf{t}}{(1-\delta_\mathbf{s})(1-\delta_\mathbf{t})} = \gamma_\mathbf{s}\gamma_\mathbf{t}, \tag{37}$$

we obtain the searched expressions for $\gamma_{\mathbf{s}\oplus\mathbf{t}}$ and $\delta_{\mathbf{s}\oplus\mathbf{t}} = \gamma_{\mathbf{s}\oplus\mathbf{t}}/(1+\gamma_{\mathbf{s}\oplus\mathbf{t}})$. •

Lemma 2.6. *For $\boldsymbol{s}$ and $\boldsymbol{t}$ in B^n,*

$$\tau(\boldsymbol{s},\boldsymbol{t}) = \tau(\boldsymbol{s}\oplus\boldsymbol{t},\boldsymbol{t}), \tag{38}$$

which means that the Einstein addition satisfies the left loop property P2.

Proof. We have to show that

$$\frac{1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st}}{(N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st}))^{\frac12}} = \frac{1+\delta_{\mathbf{s}\oplus\mathbf{t}}\delta_\mathbf{t}(\mathbf{s}\oplus\mathbf{t})\mathbf{t}}{\left(N(1+\delta_{\mathbf{s}\oplus\mathbf{t}}\delta_\mathbf{t}(\mathbf{s}\oplus\mathbf{t})\mathbf{t})\right)^{\frac12}}. \tag{39}$$

With Eq.(33), we obtain

(40)
$$1+\delta_{\mathbf{s}\oplus\mathbf{t}}\delta_\mathbf{t}(\mathbf{s}\oplus\mathbf{t})\mathbf{t} = 1+\left[\frac{N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})-N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t})}{2N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})}\right]\delta_\mathbf{t}\left[\frac{2(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t})(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{ts})\mathbf{t}}{N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})-N(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t})}\right]$$

$$= \left[\frac{(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{ts})+\delta_\mathbf{t}(\delta_\mathbf{s}\mathbf{s}+\delta_\mathbf{t}\mathbf{t})\mathbf{t}}{N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})}\right](1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st}) \tag{41}$$

$$= \lambda(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st}), \tag{42}$$

where

$$\lambda = \frac{1+2\delta_\mathbf{s}\delta_\mathbf{t}\langle\mathbf{s}|\mathbf{t}\rangle+\delta_\mathbf{t}^2\|\mathbf{t}\|^2}{N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})}, \tag{43}$$

is a scalar of $\mathbb{R}_{n,0}$. This implies

$$\frac{1+\delta_{\mathbf{s}\oplus\mathbf{t}}\delta_\mathbf{t}(\mathbf{s}\oplus\mathbf{t})\mathbf{t}}{\left(N(1+\delta_{\mathbf{s}\oplus\mathbf{t}}\delta_\mathbf{t}(\mathbf{s}\oplus\mathbf{t})\mathbf{t})\right)^{\frac12}} = \frac{\lambda(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st})}{\lambda\,(N(1+\delta_\mathbf{s}\delta_\mathbf{t}\mathbf{st}))^{\frac12}}. \bullet \tag{44}$$

We finish the proof of Theorem 2.2 by verifying that the Einstein addition satisfies the gyroassociativity property P1.

Lemma 2.7. *For $\boldsymbol{s}$, $\boldsymbol{t}$, and $\boldsymbol{u}$ in B^n,*

$$\boldsymbol{s}\oplus(\boldsymbol{t}\oplus\boldsymbol{u}) = (\boldsymbol{s}\oplus\boldsymbol{t})\oplus\left(\tau(\boldsymbol{s},\boldsymbol{t})\boldsymbol{u}\tau(\boldsymbol{s},\boldsymbol{t})^{-1}\right), \tag{45}$$

which means that the Einstein addition satisfies the gyroassociativity property P1.

Proof. Since

$$\|\tau(\mathbf{s},\mathbf{t})\mathbf{u}\tau(\mathbf{s},\mathbf{t})^{-1}\| = \|\mathbf{u}\|, \tag{46}$$

then

$$\gamma_{\tau(\mathbf{s},\mathbf{t})\mathbf{u}\tau(\mathbf{s},\mathbf{t})^{-1}} = \gamma_\mathbf{u}, \qquad \delta_{\tau(\mathbf{s},\mathbf{t})\mathbf{u}\tau(\mathbf{s},\mathbf{t})^{-1}} = \delta_\mathbf{u}. \tag{47}$$

The LHT of Eq.(45) writes

$$\mathbf{s}\oplus(\mathbf{t}\oplus\mathbf{u}) = \frac{2\left(\delta_\mathbf{s}\mathbf{s}+\delta_{\mathbf{t}\oplus\mathbf{u}}(\mathbf{t}\oplus\mathbf{u})\right)\left(1+\delta_\mathbf{s}\delta_{\mathbf{t}\oplus\mathbf{u}}(\mathbf{t}\oplus\mathbf{u})\mathbf{s}\right)}{N\left(1+\delta_\mathbf{s}\delta_{\mathbf{t}\oplus\mathbf{u}}\mathbf{s}(\mathbf{t}\oplus\mathbf{u})\right)-N\left(\delta_\mathbf{s}\mathbf{s}+\delta_{\mathbf{t}\oplus\mathbf{u}}(\mathbf{t}\oplus\mathbf{u})\right)}. \tag{48}$$

From Eq.(13) and Eq.(33), we obtain

$$\delta_{\mathbf{t} \oplus \mathbf{u}} \mathbf{t} \oplus \mathbf{u} = \frac{(\delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u})(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut})}{N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu})}. \tag{49}$$

The numerator of the RHT of Eq.(48) writes

$$\frac{2\left[(\delta_{\mathbf{s}}\mathbf{s}(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu}) + \delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u})(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut})\right]\left[(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu})(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut} + (\delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u})\delta_{\mathbf{s}}\mathbf{s})\right]}{(N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu}))^2} \tag{50}$$

$$= \frac{2\left[\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{stu}\right]\left[1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts} + \delta_{\mathbf{s}}\delta_{\mathbf{u}}\mathbf{us}\right]}{N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu})}. \tag{51}$$

Moreover,

$$N\left(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t} \oplus \mathbf{u}}\mathbf{s}(\mathbf{t} \oplus \mathbf{u})\right) = \frac{N\left[(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu})(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut} + (\delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u})\delta_{\mathbf{s}}\mathbf{s})\right]}{(N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu}))^2} \tag{52}$$

$$= \frac{N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut} + (\delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u})\delta_{\mathbf{s}}\mathbf{s})}{N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu})}, \tag{53}$$

and

$$N\left(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t} \oplus \mathbf{u}}(\mathbf{t} \oplus \mathbf{u})\right) = \frac{N\left[(\delta_{\mathbf{s}}\mathbf{s}(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu}) + \delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u})(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut})\right]}{(N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu}))^2} \tag{54}$$

$$= \frac{N(\delta_{\mathbf{s}}\mathbf{s}(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu}) + \delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u})}{N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{tu})}. \tag{55}$$

This gives

$$\mathbf{s} \oplus (\mathbf{t} \oplus \mathbf{u}) = \frac{2\left[\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{stu}\right]\left[1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts} + \delta_{\mathbf{s}}\delta_{\mathbf{u}}\mathbf{us}\right]}{N(1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts} + \delta_{\mathbf{s}}\delta_{\mathbf{u}}\mathbf{us}) - N(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{stu})}. \tag{56}$$

The RHT of Eq.(45) writes

$$(\mathbf{s} \oplus \mathbf{t}) \oplus \left(\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1}\right) \tag{57}$$

$$= \frac{2\left(\delta_{\mathbf{s} \oplus \mathbf{t}}\mathbf{s} \oplus \mathbf{t} + \delta_{\mathbf{u}}\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1}\right)\left(1 + \delta_{\mathbf{s} \oplus \mathbf{t}}\delta_{\mathbf{u}}(\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1})(\mathbf{s} \oplus \mathbf{t})\right)}{N\left(1 + \delta_{\mathbf{s} \oplus \mathbf{t}}\delta_{\mathbf{u}}(\mathbf{s} \oplus \mathbf{t})(\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1})\right) - N\left(\delta_{\mathbf{s} \oplus \mathbf{t}}\mathbf{s} \oplus \mathbf{t} + \delta_{\mathbf{u}}\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1}\right)}, \tag{58}$$

with

$$\delta_{\mathbf{s} \oplus \mathbf{t}}\mathbf{s} \oplus \mathbf{t} + \delta_{\mathbf{u}}\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1} = \frac{\left[(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}) + \delta_{\mathbf{u}}(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})\mathbf{u}\right](1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts})}{N(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})}, \tag{59}$$

and

$$1 + \delta_{\mathbf{s} \oplus \mathbf{t}}\delta_{\mathbf{u}}(\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1})(\mathbf{s} \oplus \mathbf{t}) = \frac{(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})\left[(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}) + \delta_{\mathbf{u}}\mathbf{u}\left(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}\right)\right]}{N(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})}. \tag{60}$$

Thus, the numerator of the RHT of Eq.(57) writes

$$\frac{2\left[(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}) + \delta_{\mathbf{u}}(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})\mathbf{u}\right](1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts})(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})\left[(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}) + \delta_{\mathbf{u}}\mathbf{u}\left(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}\right)\right]}{(N(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st}))^2} \tag{61}$$

$$= \frac{2\left[(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}) + \delta_{\mathbf{u}}(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})\mathbf{u}\right]\left[(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}) + \delta_{\mathbf{u}}\mathbf{u}\left(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}\right)\right]}{N(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})} \tag{62}$$

$$= \frac{2\left[\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t} + \delta_{\mathbf{u}}\mathbf{u} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{stu}\right]\left[1 + \delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut} + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts} + \delta_{\mathbf{s}}\delta_{\mathbf{u}}\mathbf{us}\right]}{N(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})}. \tag{63}$$

Moreover,

$$N(1 + \delta_{\mathbf{s} \oplus \mathbf{t}}\delta_{\mathbf{u}}(\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1})(\mathbf{s} \oplus \mathbf{t})) = \frac{N((1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}) + \delta_{\mathbf{u}}\mathbf{u}\left(\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}\right))}{N(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})}, \tag{64}$$

and

$$N(\delta_{\mathbf{s} \oplus \mathbf{t}}\mathbf{s} \oplus \mathbf{t} + \delta_{\mathbf{u}}\tau(\mathbf{s}, \mathbf{t})\mathbf{u}\tau(\mathbf{s}, \mathbf{t})^{-1}) = \frac{N((\delta_{\mathbf{s}}\mathbf{s} + \delta_{\mathbf{t}}\mathbf{t}) + \delta_{\mathbf{u}}(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})\mathbf{u})}{N(1 + \delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{st})}. \tag{65}$$

Finally,

(66)
$$(\mathbf{s}\oplus\mathbf{t})\oplus\left(\tau(\mathbf{s},\mathbf{t})\mathbf{u}\tau(\mathbf{s},\mathbf{t})^{-1}\right)=\frac{2\left[\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t}+\delta_{\mathbf{u}}\mathbf{u}+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{stu}\right]\left[1+\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut}+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}+\delta_{\mathbf{s}}\delta_{\mathbf{u}}\mathbf{us}\right]}{N(1+\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{ut}+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\mathbf{ts}+\delta_{\mathbf{s}}\delta_{\mathbf{u}}\mathbf{us})-N(\delta_{\mathbf{s}}\mathbf{s}+\delta_{\mathbf{t}}\mathbf{t}+\delta_{\mathbf{u}}\mathbf{u}+\delta_{\mathbf{s}}\delta_{\mathbf{t}}\delta_{\mathbf{u}}\mathbf{stu})}$$
(67)
$$=\mathbf{s}\oplus(\mathbf{t}\oplus\mathbf{u}).\ \bullet$$

2.3. **The hyperbolic geometry of Einstein gyrogroup.** For the sake of completeness, we detail here how to endow the interior of the unit ball B^n, for all $n\geq 2$, with a hyperbolic Riemannian metric, namely the Hilbert-Klein metric, from the Einstein gyrogroup structure (see, e.g., [22] for $n=2$, and [7] for $n=3$).

To compute the metric tensor, we consider a vector $\mathbf{v}=(v^1,\ldots,v^n)$ of $\mathbb{R}^n$ and $\mathbf{s}=(s^1,\ldots,s^n)$ in the interior of B^n. We have

$$(\mathbf{s}+\mathbf{v})\ominus\mathbf{s}=v^i\boldsymbol{\tau}_i(\mathbf{s})+o(\|\mathbf{v}\|), \tag{68}$$

where the vectors

$$\boldsymbol{\tau}_i(\mathbf{s})=\frac{\partial\left[(\mathbf{s}+\mathbf{v})\ominus\mathbf{s}\right]}{\partial v^i}|_{\mathbf{v}=\mathbf{0}} \tag{69}$$

form a basis of the tangent space $T_{\mathbf{s}}B^n$. The coefficients of the metric tensor are then given by

$$g_{ij}(\mathbf{s})=\langle\boldsymbol{\tau}_i(\mathbf{s})|\boldsymbol{\tau}_j(\mathbf{s})\rangle. \tag{70}$$

Proposition 2.8. *With the above notations, the hyperbolic Riemannian metric of the Einstein gyrogroup is the Hilbert-Klein metric given by*[1]

$$d\sigma^2(\boldsymbol{s})=g_{ij}(\boldsymbol{s})ds^ids^j=\frac{d\boldsymbol{s}^2}{(1-\|\boldsymbol{s}\|^2)}+\frac{(\boldsymbol{s}\cdot d\boldsymbol{s})^2}{(1-\|\boldsymbol{s}\|^2)^2}, \tag{71}$$

where

$$d\boldsymbol{s}^2=\sum_{i=1}^n(ds^i)^2,\qquad(\boldsymbol{s}\cdot d\boldsymbol{s})^2=\left(\sum_{i=1}^n s^ids^i\right)^2. \tag{72}$$

Proof. We have[2]

$$(\mathbf{s}+\mathbf{v})\ominus\mathbf{s}=\frac{1}{1-\langle\mathbf{s}+\mathbf{v}|\mathbf{s}\rangle}\left[(\mathbf{s}+\mathbf{v})-\frac{\mathbf{s}}{\gamma_{\mathbf{s}+\mathbf{v}}}-\frac{\gamma_{\mathbf{s}+\mathbf{v}}\langle\mathbf{s}+\mathbf{v}|\mathbf{s}\rangle(\mathbf{s}+\mathbf{v})}{1+\gamma_{\mathbf{s}+\mathbf{v}}}\right] \tag{73}$$

$$=\frac{\gamma_{\mathbf{s}+\mathbf{v}}\left[\left(1+\sqrt{1-\|\mathbf{s}+\mathbf{v}\|^2}\right)\mathbf{v}+\|\mathbf{s}+\mathbf{v}\|^2\mathbf{s}-\langle\mathbf{s}+\mathbf{v}|\mathbf{s}\rangle(\mathbf{s}+\mathbf{v})\right]}{(1-\langle\mathbf{s}+\mathbf{v}|\mathbf{s}\rangle)(1+\gamma_{\mathbf{s}+\mathbf{v}})} \tag{74}$$

$$=\frac{\left[\left(1+\sqrt{1-\|\mathbf{s}+\mathbf{v}\|^2}\right)\mathbf{v}+\|\mathbf{v}\|^2\mathbf{s}-\|\mathbf{s}\|^2\mathbf{v}+\langle\mathbf{v}|\mathbf{s}\rangle\mathbf{s}-\langle\mathbf{v}|\mathbf{s}\rangle\mathbf{v}\right]}{(1-\langle\mathbf{s}+\mathbf{v}|\mathbf{s}\rangle)\left(1+\sqrt{1-\|\mathbf{s}+\mathbf{v}\|^2}\right)}. \tag{75}$$

Consequently,

(76)
$$\boldsymbol{\tau}_i(\mathbf{s})=\frac{1}{(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)}\frac{\partial\left[\mathbf{v}\left(1+\sqrt{1-\|\mathbf{s}+\mathbf{v}\|^2}\right)+\|\mathbf{v}\|^2\mathbf{s}-\|\mathbf{s}\|^2\mathbf{v}+\langle\mathbf{v}|\mathbf{s}\rangle\mathbf{s}-\langle\mathbf{v}|\mathbf{s}\rangle\mathbf{v}\right]}{\partial v^i}|_{\mathbf{v}=\mathbf{0}}.$$

Then,

(77)
$$\frac{\partial\left[v^i\left(1+\sqrt{1-\|\mathbf{s}+\mathbf{v}\|^2}\right)+\|\mathbf{v}\|^2s^i-\|\mathbf{s}\|^2v^i+\langle\mathbf{v}|\mathbf{s}\rangle s^i-\langle\mathbf{v}|\mathbf{s}\rangle v^i\right]}{\partial v^i}|_{\mathbf{v}=\mathbf{0}}=\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)-\|\mathbf{s}\|^2+(s^i)^2,$$

[1] From now on we adopt Einstein's summation convention summing over upper and lower repeated indices.

[2] Note that, in the following $\ominus\mathbf{s}=\oplus(-\mathbf{s})$, for all $\mathbf{s}$ in $\mathbb{B}^n$.

and

$$\frac{\partial\left[v^k\left(1+\sqrt{1-\|\mathbf{s}+\mathbf{v}\|^2}\right)+\|\mathbf{v}\|^2 s^k-\|\mathbf{s}\|^2 v^k+\langle\mathbf{v}|\mathbf{s}\rangle s^k-\langle\mathbf{v}|\mathbf{s}\rangle v^k\right]}{\partial v^i}|_{\mathbf{v}=\mathbf{0}}=s^i s^k, \tag{78}$$

when $k\neq i$. Thus, the i-th component[3] of $\boldsymbol{\tau}_i(\mathbf{s})$ is

$$\boldsymbol{\tau}_i^i(\mathbf{s})=\frac{\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)-\|\mathbf{s}\|^2+(s^i)^2}{(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)}, \tag{79}$$

and its k-th component, for $k\neq i$, is

$$\boldsymbol{\tau}_i^k(\mathbf{s})=\frac{s^i s^k}{(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)}. \tag{80}$$

We obtain

$$g_{ii}(\mathbf{s})=\langle\boldsymbol{\tau}_i(\mathbf{s})|\boldsymbol{\tau}_i(\mathbf{s})\rangle=\frac{\left[\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)-\|\mathbf{s}\|^2+(s^i)^2\right]^2+(s^i)^2(\|\mathbf{s}\|^2-(s^i)^2)}{\left[(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)\right]^2} \tag{81}$$

$$=\frac{\left[\sqrt{1-\|\mathbf{s}\|^2}\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)+(s^i)^2\right]^2+(s^i)^2(\|\mathbf{s}\|^2-(s^i)^2)}{\left[(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)\right]^2} \tag{82}$$

$$=\frac{(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)^2+(s^i)^2\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)^2}{\left[(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)\right]^2} \tag{83}$$

$$=\frac{(1-\|\mathbf{s}\|^2)+(s^i)^2}{(1-\|\mathbf{s}\|^2)^2}. \tag{84}$$

We have also for $i\neq j$

(85)

$$g_{ij}(\mathbf{s})=\langle\boldsymbol{\tau}_i(\mathbf{s})|\boldsymbol{\tau}_j(\mathbf{s})\rangle=\frac{s^i s^j\left(\|\mathbf{s}\|^2-(s^i)^2-(s^j)^2+2\sqrt{1-\|\mathbf{s}\|^2}\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)+(s^i)^2+(s^j)^2\right)}{\left[(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)\right]^2}$$

$$=\frac{s^i s^j\left(\|\mathbf{s}\|^2+2\sqrt{1-\|\mathbf{s}\|^2}(1+\sqrt{1-\|\mathbf{s}\|^2})\right)}{\left[(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)\right]^2} \tag{86}$$

$$=\frac{s^i s^j\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)^2}{\left[(1-\|\mathbf{s}\|^2)\left(1+\sqrt{1-\|\mathbf{s}\|^2}\right)\right]^2} \tag{87}$$

$$=\frac{s^i s^j}{(1-\|\mathbf{s}\|^2)^2}. \tag{88}$$

Finally, combining Eq.(81) and Eq.(85), we obtain Eq.(71). •

3. Spin factors

Spin factors form "the most mysterious of the four infinite series of [simple] Jordan algebras" [2]. They were introduced for the first time under this name by Topping [21]. In this section, we first explain how the unit ball of $\mathbb{R}^n$ embeds into the state space of the spin factor $\mathbb{R}\oplus\mathbb{R}^n$, identified with a section of the future light cone in $(n+1)$-dimensional Minkowski space. Then, following

[3] Here $\boldsymbol{\tau}_i^i(\mathbf{s})$ denotes the i-th component, with no summation.

[10], we construct an embedding of this spin factor into a multipartite rebit system, realized as a tensor product of algebras of 2×2 real symmetric matrices.

3.1. **Definitions and properties.** The direct sum $\mathbb{R} \oplus \mathbb{R}^n$ equipped with the following product:

$$a \circ b = (a_0, \mathbf{a}) \circ (b_0, \mathbf{b}) = (a_0 b_0 + \langle \mathbf{a}|\mathbf{b}\rangle, b_0\mathbf{a} + a_0\mathbf{b}), \tag{89}$$

where $\langle \ | \ \rangle$ denotes the Euclidean scalar product of $\mathbb{R}^n$, is called a spin factor and is denoted $Spin(\mathbb{R}^n)$. This spin factor can be seen as the subalgebra of the Clifford algebra $\mathbb{R}_{n,0}$ (with the quadratic form given by the Euclidean scalar product) generated by the scalar unit and a spin system of vectors $\mathbf{e}_i$, with $i = 1, \cdots, n$. Indeed, the Clifford product of two elements of this subalgebra writes:

$$ab = (a_0, \mathbf{a})(b_0, \mathbf{b}) = (a_0, a^1\mathbf{e}_1 + \cdots + a^n\mathbf{e}_n)(b_0, b^1\mathbf{e}_1 + \cdots + b^n\mathbf{e}_n) \tag{90}$$

$$= (a_0, a^i\mathbf{e}_i)(b_0, b^j\mathbf{e}_j) = (a_0 b_0 + \langle \mathbf{a}|\mathbf{b}\rangle, b_0\mathbf{a} + a_0\mathbf{b}), \tag{91}$$

using the fact that, in the Clifford algebra $\mathbb{R}_{n,0}$, $\mathbf{e}_i^2 = 1$, for $i = 1, ..., n$, and $\mathbf{e}_i\mathbf{e}_j = \mathbf{e}_i \wedge \mathbf{e}_j$ for $i \neq j$ (these last products having both zero scalar and vector parts).

The spin factor $Spin(\mathbb{R}^n)$ is a Euclidean formally real Jordan algebra for the Euclidean scalar product on $\mathbb{R}^{n+1}$,

$$\langle a|b\rangle = a_0 b_0 + \langle \mathbf{a}|\mathbf{b}\rangle, \tag{92}$$

which means that

$$\langle L_a b|c\rangle = \langle b|L_a c\rangle, \tag{93}$$

where

$$L_a b = a \circ b \tag{94}$$

denotes the left multiplication.

The rank $rk(a)$ of an element $a = (a_0, \mathbf{a})$ of the spin factor $Spin(\mathbb{R}^n)$ is by definition the dimension of the associative[4] subalgebra $alg(a)$ of $Spin(\mathbb{R}^n)$ generated by a, *i.e.*

$$rk(a) := dim_{\mathbb{R}}(alg(a)). \tag{95}$$

It is therefore less or equal to $n + 1$. Clearly, if $\mathbf{a} \neq \mathbf{0}$, $(1, \mathbf{0}) = (a_0, \mathbf{a})^{\circ 0}$ and $(a_0, \mathbf{a})$ are linearly independent. Moreover,

$$(a_0, \mathbf{a})^{\circ 2} = (a_0^2 + \|\mathbf{a}\|^2, 2a_0\mathbf{a}) = 2a_0(a_0, \mathbf{a})^{\circ 1} + (\|\mathbf{a}\|^2 - a_0^2)(a_0, \mathbf{a})^{\circ 0}, \tag{96}$$

so that $rk(a) \leq 2$. If $a_0 \neq 0$, then $rk(a_0, \mathbf{0}) = 1$. The rank of the spin factor $Spin(\mathbb{R}^n)$, *i.e.* the maximal possible rank among its elements, is thus equal to 2. The elements whose rank is precisely equal to 2 are usually called regular elements.

The ring ideal of a regular element a:

$$\mathcal{I}(a) = \{P \in \mathbb{R}[X],\ P_{|alg(a)}(a) = 0\}, \tag{97}$$

is principal, generated by a unique monic polynomial of degree $rk(a)$:

$$P_a(X) = X^2 - 2a_0 X + (a_0^2 - \|\mathbf{a}\|^2), \tag{98}$$

which is by definition the minimal polynomial of the element a. Using Eq.(96), one can check that the matrix of the restriction L'_a of the linear map L_a to the subalgebra $alg(a)$ is given in the basis $\{(1, \mathbf{0}), a\}$ by

$$[L'_a] = \begin{pmatrix} 0 & \|\mathbf{a}\|^2 - a_0^2 \\ 1 & 2a_0 \end{pmatrix}. \tag{99}$$

This leads, in coherence with the expression given by Eq.(98) for the minimal polynomial of a, to introduce the following definitions (see, e.g., [12]).

Definition 3.1. The generic trace $Tr(a)$, resp. the generic determinant $Det(a)$, of an element $a = (a_0, \mathbf{a})$ of the spin factor $Spin(\mathbb{R}^n)$ is equal to $2a_0$, resp. to $a_0^2 - \|\mathbf{a}\|^2$. •

[4] Since a Jordan algebra is power associative, the subalgebra $alg(a)$ is associative and in particular is a ring.

With this definition of the trace, we have

$$Tr(a \circ b) = Tr(L_a b) = 2\langle a|b\rangle. \tag{100}$$

An element a of $Spin(\mathbb{R}^n)$ is idempotent, *i.e.* satisfies $a^{\circ 2} = a$, if and only if $a = (0, \mathbf{0})$, or $a = (1, \mathbf{0})$, or $a = (1, \mathbf{s})/2$ with $\|\mathbf{s}\| = 1$. We can decompose every element a of $Spin(\mathbb{R}^n)$ as follows:

$$a = (a_0, \mathbf{a}) = (a_0 + \|\mathbf{a}\|)s_1 + (a_0 - \|\mathbf{a}\|)s_2, \tag{101}$$

with

$$s_1 = (1, \mathbf{a}/\|\mathbf{a}\|)/2, \quad s_2 = (1, -\mathbf{a}/\|\mathbf{a}\|)/2. \tag{102}$$

The two elements s_1 and s_2 are idempotents with

$$Tr(s_1) = Tr(s_2) = 1, \quad Det(s_1) = Det(s_2) = 0, \tag{103}$$

$$\langle s_1|s_2\rangle = 0. \tag{104}$$

3.2. **State space of spin factors.** It is known that the domain of positivity of the spin factor $Spin(\mathbb{R}^n) = \mathbb{R} \oplus \mathbb{R}^n$ is the set of its squares (see, e.g., [2]), and more precisely

$$\overline{\mathcal{C}}(\mathbb{R} \oplus \mathbb{R}^n) = \{a \in \mathbb{R} \oplus \mathbb{R}^n,\ a \geq 0\} = \{a \in \mathbb{R} \oplus \mathbb{R}^n,\ \exists b \in \mathbb{R} \oplus \mathbb{R}^n,\ a = b^2\} \tag{105}$$

$$= \{a = (a_0, \mathbf{a}) \in \mathbb{R} \oplus \mathbb{R}^n,\ a_0^2 - \|\mathbf{a}\|^2 \geq 0,\ a_0 \geq 0\}. \tag{106}$$

Geometrically speaking, the domain of positivity $\overline{\mathcal{C}}(\mathbb{R} \oplus \mathbb{R}^n)$ is the future light cone of the $(n+1)$-dimensional Minkowski space.

Definition 3.2. The state space of the spin factor $Spin(\mathbb{R}^n)$ is the subset of $\overline{\mathcal{C}}(\mathbb{R} \oplus \mathbb{R}^n)$ whose elements are of generic trace equal to 1, *i.e.*

$$\mathcal{S}(\mathbb{R} \oplus \mathbb{R}^n) = \{s = (1, \mathbf{s})/2 \in \mathbb{R} \oplus \mathbb{R}^n,\ \|\mathbf{s}\| \leq 1\}. \bullet \tag{107}$$

A state $(1, \mathbf{s})/2 \in \mathcal{S}(\mathbb{R} \oplus \mathbb{R}^n)$ is a pure state if and only if it is idempotent, that is if and only if the norm of $\mathbf{s}$ is 1. We can thus consider the state space $\mathcal{S}(\mathbb{R} \oplus \mathbb{R}^n)$ as the unit ball $B^n = \{\mathbf{s} \in \mathbb{R}^n, \|\mathbf{s}\| \leq 1\}$ embedded in the domain of positivity $\overline{\mathcal{C}}(\mathbb{R} \oplus \mathbb{R}^n)$ by the map

$$\Sigma : B^n \longrightarrow \mathcal{S}(\mathbb{R} \oplus \mathbb{R}^n) \subset \overline{\mathcal{C}}(\mathbb{R} \oplus \mathbb{R}^n) \tag{108}$$

defined by $\Sigma(\mathbf{s}) = (1, \mathbf{s})/2$.

Example 3.3. The states of the spin factor $Spin(\mathbb{R}^2)$ as density matrices of the rebit.

We denote $\mathcal{M}_2(\mathbb{R})$ the (special) Jordan algebra of 2×2 matrices with real entries and $\mathcal{H}_2(\mathbb{R})$ the subalgebra of symmetric matrices of $\mathcal{M}_2(\mathbb{R})$. Let σ_1 and σ_2 be the two real Pauli matrices, *i.e.*

$$\sigma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{109}$$

and σ_0 be the identity matrix of $\mathcal{M}_2(\mathbb{R})$. The Jordan product $\sigma_1 \circ \sigma_2$ is the null 2×2 matrix, and $\sigma_1 \circ \sigma_1 = \sigma_2 \circ \sigma_2 = \sigma_0$. We consider the map

$$\chi : Spin(\mathbb{R}^2) \longrightarrow \mathcal{H}_2(\mathbb{R}), \tag{110}$$

with

$$\chi(s) = \chi((s_0, \mathbf{s})) = \chi((s_0, s^i \mathbf{e}_i)) = s_0\sigma_0 + s^1\sigma_1 + s^2\sigma_2 = \begin{pmatrix} s_0 + s^1 & s^2 \\ s^2 & s_0 - s^1 \end{pmatrix}. \tag{111}$$

One can check that[5]

$$\chi(s) \circ \chi(t) = \chi((s_0, \mathbf{s})) \circ \chi((t_0, \mathbf{t})) = \chi((s_0 t_0 + \langle \mathbf{s}|\mathbf{t}\rangle, s_0\mathbf{t} + t_0\mathbf{s})) = \chi(s \circ t). \tag{112}$$

This means that the map χ is a morphism, and in fact an isomorphism, of Jordan algebras. If $s = (1, \mathbf{s})/2$ is a state of $Spin(\mathbb{R}^2)$, then $\|\mathbf{s}\| \leq 1$ and

$$\chi((1, \mathbf{s})/2) = \frac{1}{2}\begin{pmatrix} 1 + s^1 & s^2 \\ s^2 & 1 - s^1 \end{pmatrix} \tag{113}$$

[5] The same symbol $\circ$ is used for both Jordan products of $Spin(\mathbb{R}^2)$ and $\mathcal{H}(2, \mathbb{R})$.

is a semi-positive definite symmetric matrix with trace equal to 1, *i.e.* a density matrix of the quantum system whose Hilbert space is $\mathbb{R}^2$. This system corresponds to a rebit, a real vector space version of the qubit whose Hilbert space is $\mathbb{C}^2$. Consequently, the states of the spin factor $Spin(\mathbb{R}^2)$ correspond to the states of the rebit. •

The elements s_1 and s_2 appearing in decomposition (101) are pure states of the spin factor $Spin(\mathbb{R}^n)$. Thus, every state s can be written as a convex combination of two pure states:

$$s = (1, \mathbf{s})/2 = \langle s\rangle_{s_1} s_1 + \langle s\rangle_{s_2} s_2, \tag{114}$$

where

$$\langle s\rangle_{s_1} = Tr(s_1 \circ s) = (1 + \|\mathbf{s}\|)/2, \qquad \langle s\rangle_{s_2} = Tr(s_2 \circ s) = (1 - \|\mathbf{s}\|)/2 \tag{115}$$

are the expectations of (the effect) s in the states s_1 and s_2.

We denote as before $\mathcal{M}_n(\mathbb{R})$ the (special) Jordan algebra of $n \times n$ matrices with real entries and $\mathcal{H}_n(\mathbb{R})$ the subalgebra of symmetric matrices of $\mathcal{M}_n(\mathbb{R})$. Let σ_1 and σ_2 be the two real Pauli 2×2 matrices and σ_0 be the 2×2 identity matrix. Let also $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be an orthonormal basis of the real vector space $\mathbb{R}^n$. We consider the linear map defined by (see [10])

$$\chi : Spin(\mathbb{R}^n) \longrightarrow \mathcal{H}_2(\mathbb{R})^{\otimes(n-1)} \subset \mathcal{H}_{2^{(n-1)}}(\mathbb{R}) \tag{116}$$

$$\begin{aligned}
\chi_0 &:= \chi((1, \mathbf{0})) = \sigma_0 \otimes \sigma_0 \otimes \sigma_0 \otimes \cdots \otimes \sigma_0,\\
\chi_1 &:= \chi((0, \mathbf{e}_1)) = \sigma_1 \otimes \sigma_0 \otimes \sigma_0 \otimes \cdots \otimes \sigma_0,\\
\chi_2 &:= \chi((0, \mathbf{e}_2)) = \sigma_2 \otimes \sigma_1 \otimes \sigma_0 \otimes \cdots \otimes \sigma_0,\\
\chi_3 &:= \chi((0, \mathbf{e}_3)) = \sigma_2 \otimes \sigma_2 \otimes \sigma_1 \otimes \cdots \otimes \sigma_0,\\
&\vdots\\
\chi_{n-1} &:= \chi((0, \mathbf{e}_{n-1})) = \sigma_2 \otimes \sigma_2 \otimes \cdots \sigma_2 \otimes \sigma_1,\\
\chi_n &:= \chi((0, \mathbf{e}_n)) = \sigma_2 \otimes \sigma_2 \otimes \sigma_2 \otimes \cdots \otimes \sigma_2.
\end{aligned}$$

Proposition 3.4. *The linear map χ is a morphism of Jordan algebras, i.e.*

$$\chi((s_0, \boldsymbol{s}) \circ (t_0, \boldsymbol{t})) = \chi((s_0, \boldsymbol{s})) \circ \chi((t_0, \boldsymbol{t})). \tag{117}$$

Proof. Since the Jordan product $\sigma_1 \circ \sigma_2$ is the null 2×2 matrix, and $\sigma_1 \circ \sigma_1 = \sigma_2 \circ \sigma_2 = \sigma_0$, we have

$$\chi_i^{\circ 2} = \chi_0, \tag{118}$$

for all $i = 1, \ldots, n$, and the products $\chi_i \circ \chi_j$ are the null matrix whenever $i \neq j$. Consequently,

$$\chi((s_0, \mathbf{s})) \circ \chi((t_0, \mathbf{t})) = (s_0 t_0 + \langle \mathbf{s}, \mathbf{t}\rangle)\chi_0 + t_0 s^i \chi_i + s_0 t^j \chi_j \tag{119}$$

$$= \chi((s_0 t_0 + \langle \mathbf{s}, \mathbf{t}\rangle, t_0 \mathbf{s} + s_0 \mathbf{t})) = \chi((s_0, \mathbf{s}) \circ (t_0, \mathbf{t})). \; \bullet \tag{120}$$

Example 3.5. The states of the spin factor $Spin(\mathbb{R}^3)$ as density matrices.

We have already seen that the states of the spin factor $Spin(\mathbb{R}^2)$ correspond to density matrices of the rebit. In the same way, the states of the spin factor $Spin(\mathbb{R}^3)$ correspond to density matrices of the composite system $\mathcal{H}_2(\mathbb{R}) \otimes \mathcal{H}_2(\mathbb{R})$. The map $\chi : Spin(\mathbb{R}^3) \longrightarrow \mathcal{H}_2(\mathbb{R}) \otimes \mathcal{H}_2(\mathbb{R})$ is given by

$$\chi((s_0, s^1\mathbf{e}_1 + s^2\mathbf{e}_2 + s^3\mathbf{e}_3)) = \begin{pmatrix} s_0 + s^1 & 0 & s^2 & s^3 \\ 0 & s_0 + s^1 & s^3 & -s^2 \\ s^2 & s^3 & s_0 - s^1 & 0 \\ s^3 & -s^2 & 0 & s_0 - s^1 \end{pmatrix}. \tag{121}$$

If $s = (1, \mathbf{s})/2$, with $\|\mathbf{s}\| \leq 1$, is a state, then

$$\chi(s) = \frac{1}{2} \begin{pmatrix} 1 + s^1 & 0 & s^2 & s^3 \\ 0 & 1 + s^1 & s^3 & -s^2 \\ s^2 & s^3 & 1 - s^1 & 0 \\ s^3 & -s^2 & 0 & 1 - s^1 \end{pmatrix}. \tag{122}$$

Note that the generic trace of $\chi(s)$ is equal to 1, although its usual trace is 2. •

4. Lüders measurements

This section is devoted to the main result of this work. We prove that Lüders operations performed on density matrices of the multipartite rebit system $\mathcal{H}_2(\mathbb{R})^{\otimes(n-1)}$ induce a gyrogroup morphism between the unit ball of $\mathbb{R}^n$ equipped with Einstein addition and the gyrogroup of density matrices equipped with the post-measurement state operation, see Theorem 4.3.

4.1. Definitions and statement of the main result.

Let

$$\chi : Spin(\mathbb{R}^n) \longrightarrow \mathcal{H}_2(\mathbb{R})^{\otimes(n-1)} \subset \mathcal{H}_{2^{(n-1)}}(\mathbb{R}) \tag{123}$$

be given by Eq.(116), and let

$$\rho_{\mathbf{s}} = \frac{1}{2}(\chi_0 + s^i\chi_i) \tag{124}$$

be the density matrix image by χ of the spin factor state $s = (1, \mathbf{s})/2$, with $\mathbf{s} = s^i\mathbf{e}_i$ and $\|\mathbf{s}\| \leq 1$. Lüders measurements are quantum measurements performed on states by means of effects (see, e.g., [11] or [3] for complements).

Definition 4.1. An effect of the spin factor is a positive element $e = (e_0, e^i\mathbf{e}_i)$ of $Spin(\mathbb{R}^n)$ such that $(1, \mathbf{0}) - e$ is also positive. In particular, $e = e_0(1, \mathbf{e})$, where $0 < e_0 \leq 1$, $\mathbf{e} = \underline{e}^i\mathbf{e}_i$, with $\underline{e}^i = e^i/e_0$, $i = 1, ..., n$, and $\|\mathbf{e}\| \leq 1$. •

We denote

$$\eta_{e_0,\mathbf{e}} = 2e_0\rho_{\mathbf{e}} \tag{125}$$

the image of the effect e by the linear map χ. Given a state s and an effect e, the outcome of the Lüders measurement performed by e on s is the generalized density matrix

$$\psi_e(s) = \eta_{e_0,\mathbf{e}}^{1/2}\rho_{\mathbf{s}}\eta_{e_0,\mathbf{e}}^{1/2} = \langle e\rangle_s\varphi_e(s), \tag{126}$$

where $\langle e\rangle_s$ is the expectation of the measurement, and $\varphi_e(s)$ is the state-update density matrix of the post-measurement state. The square root of $\eta_{e_0,\mathbf{e}}$ is the Kraus operator associated to the measurement, and by definition of an effect, the expectation satisfies $0 \leq \langle e\rangle_s \leq 1$.

The goal of this section is to prove the following theorem.

Theorem 4.2. *Let s be a state and e be an effect of the spin factor $Spin(\mathbb{R}^n)$. The outcome of the Lüders measurement performed by e on s is given by*

$$\psi_e(s) = \frac{e_0}{2}\left(1 + \langle \boldsymbol{e}|\boldsymbol{s}\rangle\right)\left(\chi_0 + (\boldsymbol{e} \oplus \boldsymbol{s})^i\chi_i\right), \tag{127}$$

where

$$\boldsymbol{e} \oplus \boldsymbol{s} = (\boldsymbol{e} \oplus \boldsymbol{s})^i\boldsymbol{e}_i \tag{128}$$

is the Einstein sum of the effect vector $\mathbf{e}$ *with the state vector* $\boldsymbol{s}$. •

Note that

$$\varphi_e(s) = \frac{1}{2}\left(\chi_0 + (\mathbf{e} \oplus \mathbf{s})^i\chi_i\right) = \rho_{\mathbf{e} \oplus \mathbf{s}}, \tag{129}$$

and let us denote

$$\mathcal{D} = \chi\left(\mathcal{S}(\mathbb{R} \oplus \mathbb{R}^n)\right) = \chi \circ \Sigma(B^n) \tag{130}$$

the image by χ of the spin factor state space. As an immediate consequence of Theorem 4.2 we obtain the following theorem.

Theorem 4.3. *The map $\chi\circ\Sigma$ provides an isomorphism of gyrogroups between the ball B^n equipped with the Einstein addition $(\boldsymbol{e},\boldsymbol{s})\longmapsto \boldsymbol{e}\oplus\boldsymbol{s}$, and $\mathcal{D}$ equipped with the post-measurement state operation*

$$\rho_{\boldsymbol{e}}\oplus\rho_{\boldsymbol{s}}=\frac{2\rho_{\boldsymbol{e}}^{1/2}\rho_{\boldsymbol{s}}\rho_{\boldsymbol{e}}^{1/2}}{(1+\langle\boldsymbol{e}|\boldsymbol{s}\rangle)}=\rho_{\boldsymbol{e}\oplus\boldsymbol{s}}.\ \bullet \tag{131}$$

4.2. **Proof of Theorem 4.2.** With Eq.(124) and Eq.(125), we have

$$\psi_e(s)=2e_0\rho_{\mathbf{e}}^{1/2}\rho_{\mathbf{s}}\rho_{\mathbf{e}}^{1/2}=e_0\left(\rho_{\mathbf{e}}+\rho_{\mathbf{e}}^{1/2}s^i\chi_i\rho_{\mathbf{e}}^{1/2}\right). \tag{132}$$

The spin factor state $(1,\mathbf{e})/2$ is a pure state if and only if it is idempotent and, because χ is a morphism of Jordan algebras, if and only if $\rho_{\mathbf{e}}^2=\rho_{\mathbf{e}}$. This means that

$$\|\mathbf{e}\|=1\Longleftrightarrow\rho_{\mathbf{e}}^2=\rho_{\mathbf{e}}. \tag{133}$$

When $(1,\mathbf{e})/2$ is not a pure state, we have

$$\rho_{\mathbf{e}}^{1/2}=\chi((a_0,\mathbf{a})), \tag{134}$$

with the element $(a_0,\mathbf{a})$ of the spin factor $Spin(\mathbb{R}^n)$ satisfying $(a_0,\mathbf{a})^{\circ 2}=(1,\mathbf{e})/2$, that is

$$\begin{cases} 2(a_0^2+\|\mathbf{a}\|^2)=1\\ 4a_0\mathbf{a}=\mathbf{e}.\end{cases} \tag{135}$$

Recall from Eq.(11) that

$$\delta_{\mathbf{e}}^2\|\mathbf{e}\|^2=2\delta_{\mathbf{e}}-1, \tag{136}$$

or equivalently that

$$\frac{1+\delta_{\mathbf{e}}^2\|\mathbf{e}\|^2}{4\delta_{\mathbf{e}}}=\frac{1}{4\delta_{\mathbf{e}}}+\frac{\delta_{\mathbf{e}}\|\mathbf{e}\|^2}{4}=\frac{1}{2}. \tag{137}$$

Thus,

$$a_0=\frac{1}{2\sqrt{\delta_{\mathbf{e}}}},\qquad \mathbf{a}=\frac{\sqrt{\delta_{\mathbf{e}}}\mathbf{e}}{2} \tag{138}$$

solves the system, and since

$$a_0^2-\|\mathbf{a}\|^2=\frac{2(1-\delta_{\mathbf{e}})}{4\delta_{\mathbf{e}}}>0, \tag{139}$$

with $a_0>0$, the element $(a_0,\mathbf{a})$ is positive. This means that the unique positive square root of $\rho_{\mathbf{s}}$ is given by

$$\rho_{\mathbf{e}}^{1/2}=\chi\left(\left(\frac{1}{2\sqrt{\delta_{\mathbf{e}}}},\frac{\sqrt{\delta_{\mathbf{e}}}\mathbf{e}}{2}\right)\right)=\sqrt{\delta_{\mathbf{e}}}\left(\chi\left(\left(\frac{1}{2\gamma_{\mathbf{e}}},0\right)\right)+\chi((1,\mathbf{e})/2)\right) \tag{140}$$

$$=\sqrt{\delta_{\mathbf{e}}}\left(\frac{1}{2\gamma_{\mathbf{e}}}\chi_0+\rho_{\mathbf{e}}\right). \tag{141}$$

To prove Theorem 4.2 we distinguish the two cases: the state $(1,\mathbf{e})/2$, with $\mathbf{e}=\underline{e}^i\mathbf{e}_i$, is pure, and it is not pure. When $\|\mathbf{e}\|=1$, then $\mathbf{e}\oplus\mathbf{s}=\mathbf{e}$. Since $\rho_{\mathbf{e}}^2=\rho_{\mathbf{e}}$, we have to prove that

$$\psi_e(s)=e_0\left(\rho_{\mathbf{e}}+\rho_{\mathbf{e}}s^i\chi_i\rho_{\mathbf{e}}\right)=\frac{e_0}{2}\left(1+\langle\mathbf{e}|\mathbf{s}\rangle\right)\left(\chi_0+\underline{e}^j\chi_j\right)=e_0\left(1+\langle\mathbf{e}|\mathbf{s}\rangle\right)\rho_{\mathbf{e}}. \tag{142}$$

Note that in this case $\rho_{\mathbf{e}}$ is not invertible. When $\|\mathbf{e}\|<1$, we have to prove that

$$\psi_e(s)=e_0\left(\rho_{\mathbf{e}}+\rho_{\mathbf{e}}^{1/2}s^i\chi_i\rho_{\mathbf{e}}^{1/2}\right)=\frac{e_0}{2}\left(1+\langle\mathbf{e}|\mathbf{s}\rangle\right)\left(\chi_0+(\mathbf{e}\oplus\mathbf{s})^i\chi_i\right), \tag{143}$$

where $\rho_{\mathbf{e}}^{1/2}$ is given by Eq.(140). We begin with a technical lemma.

Lemma 4.4. *Let χ_i, $i=0,\dots,n$, be the matrices given by Eq.(116), and let $\boldsymbol{e}=\underline{e}^k\chi_k$, $\boldsymbol{s}=s^i\chi_i$, be two vectors of the ball B^n, we have*

$$\underline{e}^k\chi_k s^i\chi_i\underline{e}^l\chi_l=2\langle\boldsymbol{e}|\boldsymbol{s}\rangle\underline{e}^k\chi_k-\|\boldsymbol{e}\|^2s^i\chi_i. \tag{144}$$

Proof. Recall that $\chi_i^2 = \chi_0$ and $\chi_i\chi_j = -\chi_j\chi_i$, for $i \neq j$. We have[6]

$$\underline{e}^k\chi_k s^i\chi_i\underline{e}^l\chi_l = (\underline{e}^i)^2 s^i\chi_i + \sum_i \underline{e}^i s^i \underline{e}^{\widehat{i}}\chi_{\widehat{i}} + \sum_i \underline{e}^{\widehat{i}} s^i \underline{e}^i\chi_{\widehat{i}} - \sum_k (\underline{e}^k)^2 s^{\widehat{k}}\chi_{\widehat{k}} + \underline{e}^k s^i \underline{e}^l \chi_k\chi_i\chi_{l_{|k\neq i, i\neq l, l\neq k}} \tag{145}$$

$$= (\underline{e}^i)^2 s^i\chi_i + 2\sum_i \underline{e}^{\widehat{i}} s^i \underline{e}^i \chi_{\widehat{i}} - \sum_k (\underline{e}^k)^2 s^{\widehat{k}}\chi_{\widehat{k}} \tag{146}$$

$$= 2\langle \mathbf{e}|\mathbf{s}\rangle \underline{e}^k\chi_k - \|\mathbf{e}\|^2 s^i\chi_i. \bullet \tag{147}$$

We consider now the case when the spin factor state $(1,\mathbf{e})/2$ is a pure state.

Lemma 4.5. *Let $e = e_0(1,\boldsymbol{e})$ be an effect of the spin factor $Spin(\mathbb{R}^n)$, with $\boldsymbol{e} = \underline{e}^i\boldsymbol{e}_i$, $\underline{e}^i = e^i/e_0$, for $i = 1,...,n$, and $\|\mathbf{e}\| = 1$, and let $s = (1,\boldsymbol{s})/2$ be a state with $\boldsymbol{s} = s^i\boldsymbol{e}_i$, we have*

$$\psi_e(s) = e_0\left(1 + \langle \mathbf{e}|\boldsymbol{s}\rangle\right)\rho_{\boldsymbol{e}}. \tag{148}$$

Proof. According to Eq.(142), we compute

$$\rho_{\mathbf{e}} + \rho_{\mathbf{e}} s^i\chi_i\rho_{\mathbf{e}} = \frac{1}{2}(\chi_0 + \underline{e}^j\chi_j) + \frac{1}{2}(\chi_0 + \underline{e}^k\chi_k)s^i\chi_i\frac{1}{2}(\chi_0 + \underline{e}^l\chi_l) \tag{149}$$

$$= \frac{1}{2}\left(\chi_0 + \underline{e}^j\chi_j + \frac{1}{2}(s^i\chi_i + s^i\chi_i\underline{e}^l\chi_l + \underline{e}^k\chi_k s^i\chi_i + \underline{e}^k\chi_k s^i\chi_i\underline{e}^l\chi_l)\right) \tag{150}$$

$$= \frac{1}{2}\left(\chi_0 + \underline{e}^j\chi_j + \frac{1}{2}(s^i\chi_i + 2\langle \mathbf{e}|\mathbf{s}\rangle\chi_0 + \underline{e}^k\chi_k s^i\chi_i\underline{e}^l\chi_l)\right). \tag{151}$$

Since $\|\mathbf{e}\| = 1$, we obtain with Lemma 4.4

$$\rho_{\mathbf{e}} + \rho_{\mathbf{e}} s^i\chi_i\rho_{\mathbf{e}} = \frac{1}{2}\left(\chi_0 + \underline{e}^j\chi_j + \langle \mathbf{e}|\mathbf{s}\rangle\chi_0 + \langle \mathbf{e}|\mathbf{s}\rangle\underline{e}^k\chi_k\right) \tag{152}$$

$$= \frac{1}{2}\left(1 + \langle \mathbf{e}|\mathbf{s}\rangle\right)\left(\chi_0 + \underline{e}^j\chi_j\right) = \left(1 + \langle \mathbf{e}|\mathbf{s}\rangle\right)\rho_{\mathbf{e}}. \bullet \tag{153}$$

Finally, to prove Theorem 4.2, it remains to consider the case when the state $(1,\mathbf{e})/2$ is not a pure state.

Lemma 4.6. *Let $e = e_0(1,\boldsymbol{e})$ be an effect of the spin factor $Spin(\mathbb{R}^n)$, with $\boldsymbol{e} = \underline{e}^i\boldsymbol{e}_i$, $\underline{e}^i = e^i/e_0$, for $i = 1,...,n$, and $\|\mathbf{e}\| < 1$, and let $s = (1,\boldsymbol{s})/2$ be a state with $\boldsymbol{s} = s^i\boldsymbol{e}_i$, we have*

$$\psi_e(s) = \frac{e_0}{2}\left(1 + \langle \boldsymbol{e}|\boldsymbol{s}\rangle\right)\left(\chi_0 + (\boldsymbol{e} \oplus \boldsymbol{s})^i\chi_i\right). \tag{154}$$

Proof. According to Eq.(143), we compute

$$\rho_{\mathbf{e}} + \rho_{\mathbf{e}}^{1/2} s^i\chi_i\rho_{\mathbf{e}}^{1/2} = \frac{1}{2}\left(\chi_0 + \underline{e}^j\chi_j + \frac{\delta_{\mathbf{e}}}{2}\left(\frac{1}{\delta_{\mathbf{e}}}\chi_0 + \underline{e}^k\chi_k\right)s^i\chi_i\left(\frac{1}{\delta_{\mathbf{e}}}\chi_0 + \underline{e}^l\chi_l\right)\right) \tag{155}$$

$$= \frac{1}{2}\left(\chi_0 + \underline{e}^j\chi_j + \frac{\delta_{\mathbf{e}}}{2}\left(\frac{1}{\delta_{\mathbf{e}}^2}s^i\chi_i + \frac{1}{\delta_{\mathbf{e}}}s^i\chi_i\underline{e}^l\chi_l + \frac{1}{\delta_{\mathbf{e}}}\underline{e}^k\chi_k s^i\chi_i + \underline{e}^k\chi_k s^i\chi_i\underline{e}^l\chi_l\right)\right). \tag{156}$$

With Lemma 4.4, this gives

$$\rho_{\mathbf{e}} + \rho_{\mathbf{e}}^{1/2} s^i\chi_i\rho_{\mathbf{e}}^{1/2} = \frac{1}{2}\left(\chi_0 + \underline{e}^j\chi_j + \frac{1}{2}\left(\frac{1}{\delta_{\mathbf{e}}}s^i\chi_i + 2\langle \mathbf{e}|\mathbf{s}\rangle\chi_0 + \delta_{\mathbf{e}}\left(2\langle \mathbf{e}|\mathbf{s}\rangle\underline{e}^k\chi_k - \|\mathbf{e}\|^2 s^i\chi_i\right)\right)\right) \tag{157}$$

$$= \frac{1}{2}\left(1 + \langle \mathbf{e}|\mathbf{s}\rangle\right)\left(\chi_0 + \frac{1}{1 + \langle \mathbf{e}|\mathbf{s}\rangle}\left(\underline{e}^j\chi_j + \frac{1}{2}\left(\frac{1}{\delta_{\mathbf{e}}} - \delta_{\mathbf{e}}\|\mathbf{e}\|^2\right)s^i\chi_i + \delta_{\mathbf{e}}\langle \mathbf{e}|\mathbf{s}\rangle\underline{e}^k\chi_k\right)\right). \tag{158}$$

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[6] Here $\underline{e}^{\widehat{i}}\chi_{\widehat{i}}$ means Einstein's summation when the summation index runs through the set $\{1,...,n\} \setminus \{i\}$, and $\underline{e}^k s^i \underline{e}^l \chi_k\chi_i\chi_{l_{|k\neq i, i\neq l, l\neq k}}$ means Einstein's summation performed on triplet of indices (k,i,l) with k, i, and l distinct in pairs.

Since

$$\frac{1}{\delta_{\mathbf{e}}} - \delta_{\mathbf{e}}\|\mathbf{e}\|^2 = 2\left(\frac{1}{\delta_{\mathbf{e}}} - 1\right) = \frac{2}{\gamma_{\mathbf{e}}}, \tag{159}$$

we obtain

$$\rho_{\mathbf{e}} + \rho_{\mathbf{e}}^{1/2} s^i \chi_i \rho_{\mathbf{e}}^{1/2} = \frac{1}{2}\left(1 + \langle \mathbf{e}|\mathbf{s}\rangle\right)\left(\chi_0 + \frac{1}{1 + \langle \mathbf{e}|\mathbf{s}\rangle}\left(\underline{e}^j \chi_j + \frac{1}{\gamma_{\mathbf{e}}} s^i \chi_i + \delta_{\mathbf{e}} \langle \mathbf{e}|\mathbf{s}\rangle \underline{e}^k \chi_k\right)\right) \tag{160}$$

$$= \frac{1}{2}\left(1 + \langle \mathbf{e}|\mathbf{s}\rangle\right)\left(\chi_0 + (\mathbf{e} \oplus \mathbf{s})^i \chi_i\right). \bullet \tag{161}$$

5. **Conclusion**

In this work, we have shown that the unit ball of $\mathbb{R}^n$ equipped with the Einstein gyrogroup structure, embeds via a gyrogroup morphism into a gyrogroup of density matrices associated with a multipartite rebit system. This provides a unified algebraic and geometric framework connecting relativistic velocity addition with quantum measurement in spin factor state spaces, and yields a hyperbolic geometric interpretation that extends the familiar Bloch ball picture to arbitrary dimensions.

Finally, we note that gyrogroups and their associated hyperbolic geometries have recently attracted attention in the context of representation learning and hyperbolic neural networks. Hierarchical data sets often exhibit hyperbolic structures closely related to tree-like geometries in the sense of Gromov [9]; see for instance [13]. This insight has led to the design of hyperbolic neural networks, where training data are embedded in the Poincaré ball and basic operations are expressed using the Möbius gyrogroup (see [16], [17], [20]). From this perspective, the Einstein gyrogroup, together with its Hilbert-Klein geometry, may provide a mathematically natural setting for interpreting certain hyperbolic learning models within the formalism of quantum information theory. Exploring such connections at the algorithmic level is left for future work.

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